Dibyadarshini Maharatha

M.Tech – Mathematics Computation — National Institute of Technology, Delhi
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Career Objective

Aspiring Embedded Systems Engineer with a strong foundation in microcontroller programming, real-time systems, and software development. Skilled in C, Python, and embedded firmware design. Passionate about developing efficient and scalable embedded solutions. Seeking opportunities to contribute innovative solutions in embedded technology and IoT applications.

Technical Skills

- **Programming:** C, Python, C++, Embedded C
- **Embedded Systems:** PIC, ARM, STM32,AVR
- **Communication Protocols:** UART, I2C, SPI, CAN
- **Development Tools:** MPLAB X IDE, Arduino IDE, Keil, STM32CubeIDE
- **Data & ML Skills:** Machine Learning, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook
- **Algorithms Data Structures:** Sorting, Searching, Trees, Graphs, Dynamic Programming
- **Operating Systems:** Linux, Embedded Linux

Education

M.Tech – Mathematics Computation(2026)
National Institute of Technology, Delhi — CGPA:7.38
B.Tech – Electronics Telecommunication Engineering (2021)
Biju Patnaik University of Technology — CGPA: 8.70

Projects

Car Black Box using PIC Microcontroller

- Developed a vehicle monitoring system to record parameters like speed, impact force, and location.
- Integrated sensors for real-time data logging using UART communication and SD card storage.
- Implemented crash detection with impact force analysis and GPS tracking.

Technologies: PIC Microcontroller, Embedded C, SD Card, Accelerometer, GPS.

Steganography using C

- Implemented an image-based steganography algorithm to hide secret messages within image files.
- Developed an encoding and decoding mechanism using least significant bit (LSB) modification.
- Optimized file handling operations for better performance in large-scale data hiding.

Technologies: C, Bit Manipulation, File Handling, Image Processing Algorithms.

ImageRecog (Class Presenter)

- Built a interactive jupyter notebooks with embedded code cells for real-time demonstrations in class.
  Implement a deeplearning model using Tesnorflow,achieving a 76% accuracy rate in object recognition.
- Incorporated error analysis and confusion matrix visualization in presentations,aiding classmates to
- understand model limitations and area for improvement.
- Created data visualization using matplotlib to illustrate the projects dataset distribution.

Technologies: Python, TensorFlow, Matplotlib, Jupyter Notebook

Certifications Achievements

- Advanced Embedded Systems Certification (Emertxe, Govt. of India Certified)
- GATE 2024 Rank: AIR 3371 (ECE)
- 1st Prize – Intra-college Technical Quiz Circuit Design Event
- Successfully completed ”Embedded Linux and Device Drivers” training

Hereby declared that the details furnished above are true and correct to the best of knowledge and belief.